REAR-IR1: rear docking IR board. 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket):
1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

CLIFF
File: CLIFF.kicad_sch

MAIN BRUSH MOTOR
File: MAIN-BRUSH-MOTOR .kicad_sch

WHEEL MOTORs
File: WHEEL-MOTORs .kicad_sch

MAIN FAN
File: MAIN-FAN .kicad_sch

SIDE BRUSH MOTORs
File: SIDE-BRUSH-MOTORs .kicad_sch

MCU-STM32
File: STM32G473.kicad_sch

SIDE-R1
ZX-SH1_0-7PWT

CARPET-SENSOR
File: Carpet-sensor.kicad_sch

WATCHDOG
File: WATCHDOG.kicad_sch

LiDAR
File: LiDAR.kicad_sch

FRONT SENSORS
File: Front-Sensors.kicad_sch

HALL-BIN1
HALL-TOWER1
REAR-IR1
ZX-SH1_0-3PLT
ZX-SH1_0-4PLT
R215 47k
R216 47k

IMU
File: IMU-ICM-4267-P .kicad_sch

BUTTON-LEDs
File: BUTTON-LEDs .kicad_sch

POWER CONTROL
File: SYSTEM-POWER.kicad_sch

BATTERY CHARGER
File: Battery-Charger.kicad_sch

R6 2.2k
R5 2.2k

Sheet: /	
File: Main.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.4	Id: 1/15

5V USB, SENSORS, LIDAR POWER RAIL
– set 5V_EN HIGH to enable

TPS54560B (U21, U22):
– non-synchronous
– COMP 4.7k + 47nF (all-ceramic Cout, no pole cap)
– SS56 catch diode; Fsw 418 kHz (RT 232k)

VBAT-14V4
PWR_FLAG
VSENS_5V_USB_END
C119 100nF/50V
10 R146
U22
TPS54560BDDAR
BOOT
VIN
EN
RT/CLK
PAD
SW
GND
COMP
FB
L3
CYA0630-3_3UH
D43
SS56
VSENS-5V-USB
D46
YLED0603B
C115 10uF/50V
C116 10uF/50V
C117 100nF/50V
R206 232k
R168 100k
R147 4.7k
C120 47nF/50V
R148 14k
R149 2.2k
C121 10uF/10V
C122 10uF/10V
C147 10uF/10V
1k R89

VBAT-14V4
Q1
AO4407C
PWR_FLAG
VMOT-14V4
R165 10k
R36 10
C62 10uF/25V
D37
SS34
VMCU-3V3
100k R184
Q19
AO3400
WD_OK
VMOT-14V4-EN

MOTOR POWER RAIL 14.4V

– on/off motor power, no soft ramp
– set VMOT-14V4-EN LOW to enable power
  – floating (MCU flash/reset) disables power
– WD_OK LOW shuts power off

VMOT-14V4 AO4407C
– ~5 A normal load is ~0.3–0.4 W
– short stall peaks (~13 A) fine with copper under the drain
– Vgs +/–25 V, no gate divider needed

VBAT-14V4
Q13
AO4407C
POWER-EN
R67 10k
C14 1uF/25V
R171 100k
PWR-SW-G
R16 100k
POWER-BTN
C22 1uF/25V
VMCU-3V3
R173 10k
PWR-BTN-SENSE
D26
1N4148
STM-PWR-CTRL
R172 10k
R190 1k
Q23
AO3400
PWR-LATCH

STM-PWR-CTRL need to use for both EN and OFF from STM32 GPIO pin.
1– Hold EN to HIGH after buttone pressed
2– OFF board

VMCU-3V3
R193 1M
R191 100k
R192 47k
Q29
AO3400
C172 10uF/10V
U23
74LVC1G17
GND VCC
C173 100nF/50V
Q30
AO3400

Hardware off timer: holding the power button for ~4–9 s turns the MCU rail off even if the firmware hangs (Q29 releases C172, U23 trips, Q30 pulls PWR-LATCH low), Firmware turns off at 3 s.
While charging, the rail-hold (Q31 on BATTERY-CHARGER, from DC_IN) keeps the MCU rail on regardless, and charger power alone turns the MCU rail on (Q31 pulls PWR-SW-G low).

3.3V MCU POWER RAIL

– POWER_EN is 14.4V
– set POWER_EN HIGH to enable

TPS54560B (U21, U22):
– non-synchronous
– COMP 4.7k + 47nF (all-ceramic Cout, no pole cap)
– SS56 catch diode: fsw 418 kHz (RT 232k)
– EN abs max 8.4 V: R207/R208 on U21

VBAT-14V4
R207 100k
R208 22.1k
C111 100nF/50V
10 R141
U21
TPS54560BDDAR
L2
CYA0630-3_3UH
D42
SS56
PWR_FLAG
VMCU-3V3
D44
YLED0603B
1k R14
R205 232k
C112 4.7nF/50V
R143 47nF/50V
R144 100k
R145 22k
C106 10uF/50V
C95 10uF/50V
C102 100nF/50V
C113 10uF/10V
C114 10uF/10V
C146 10uF/10V

14.5V ULTRASONIC CARPET SENSOR POWER

L4
XRNR4030-10UH_M
D4
SS34_C8678
PWR_FLAG
VUSONIC-14V4
VSENS-5V-USB
U5
MT3608
NC
IN
EN
SW
GND
FB
R160 232k
R159 10k
C149 10uF/10V
C148 10uF/10V
C150 10uF/25V
C151 10uF/25V

VUSONIC-14V4 = 0.6V x (1 + 232k/10k) ~= 14.5V

VMCU-3V3
Q16
AO3401
VSENS-3V3
PWR_FLAG
100k R99
1uF/25V C176
D45
YLED0603B
1k R98
VSENS_3V3_EN
10k R195
C64 100nF/50V

3.3V SENSORS POWER RAIL

– C176 keeps Q16 off when power switches ON
– C64 soft-ramps power
– set VSENS_3V3_EN LOW to enable sensors power

MCU VBAT
– Low quiescent current 2.5µA
– Zener diode shuts off when main battery is low

VBAT-14V4
D2
ZMM9V1_C2891750
U16
HT7533-1_C14289
Vout
Vin
GND
PWR_FLAG
VMCU-BAT-3V3
C140 10uF/25V
C141 10uF/25V

PCB mounting standoffs (Wurth WA-SMSI, M3), moved here from the removed M.2 sheet.

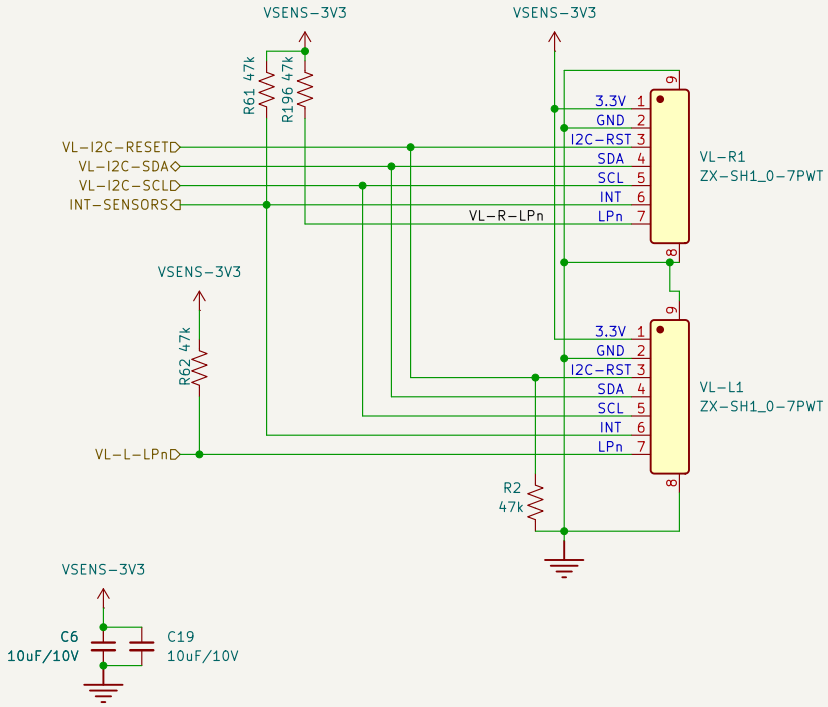
MP1 MP2 MP3 MP4 MP5 MP6

Sheet: /POWER/
File: SYSTEM-POWER.kicad_sch

Title:

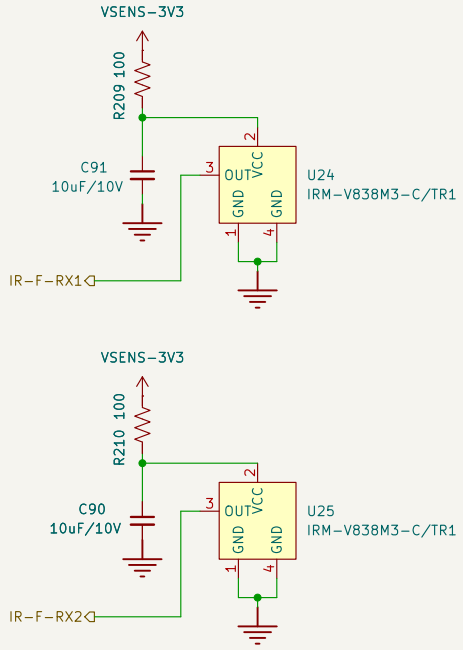
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KiCad E.D.A. 10.0.4 | Id: 2/15

2x VL53L7 ToF BOARDS



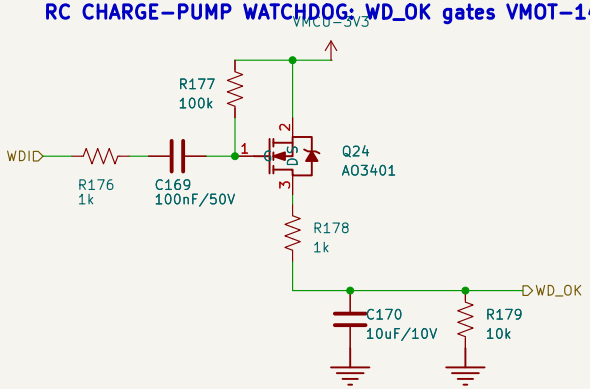
2 IR RECEIVERS

U24 left, U25 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window. ~ 90 deg FoV



Sheet: /FRONT SENSORS/	
File: Front-Sensors.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 4/15

RC CHARGE-PUMP WATCHDOG: WD_OK gates VMOT-14V4, VM-5V-LIDAR and VM-5V-WATER-PUMP

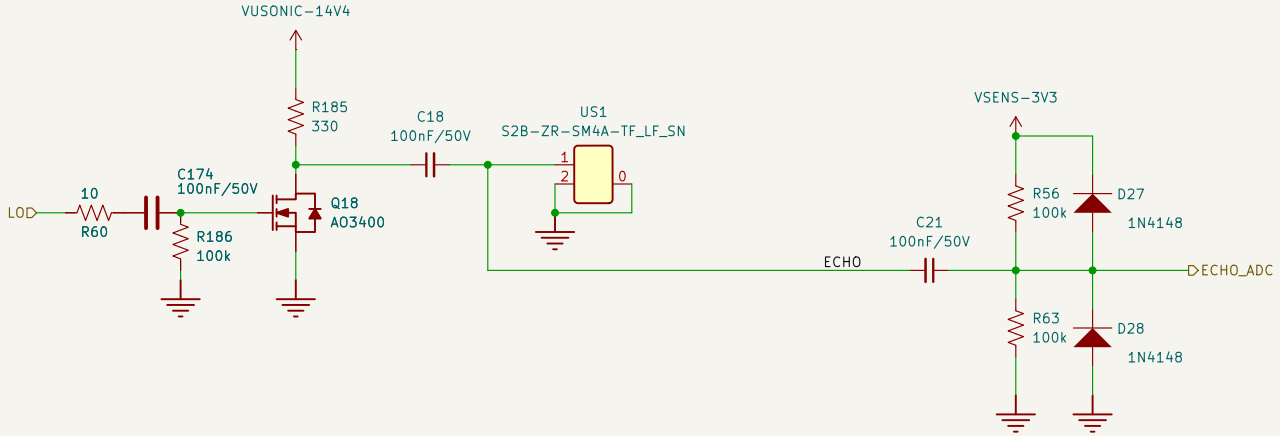


Firmware toggles WDI in software (50–1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, water pump, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down. All parts reuse existing BOM lines.

Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 5/15

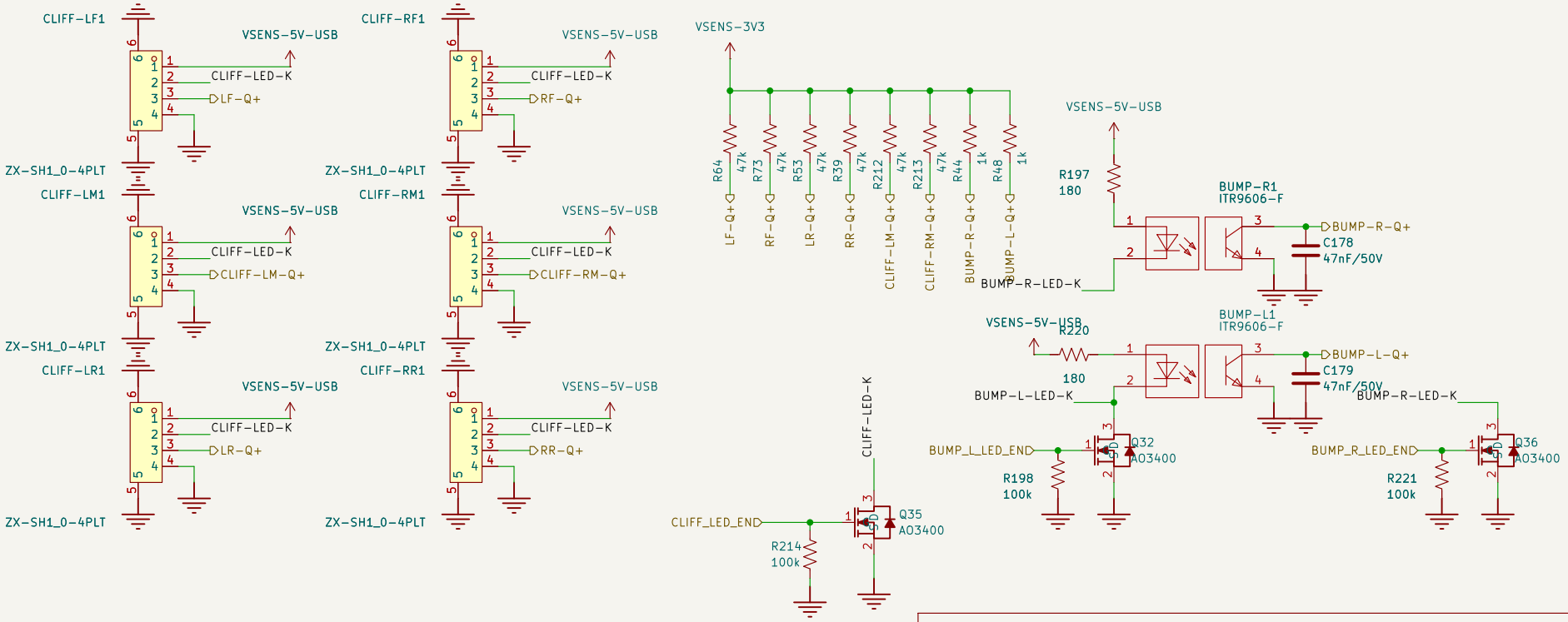
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms)
and Q18 turns off, so R185 is never stressed by firmware holding LO high.



Sheet: /CARPET-SENSOR/		
File: Carpet-sensor.kicad_sch		
Title:		
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KiCad E.D.A. 10.0.4		Id: 6/15

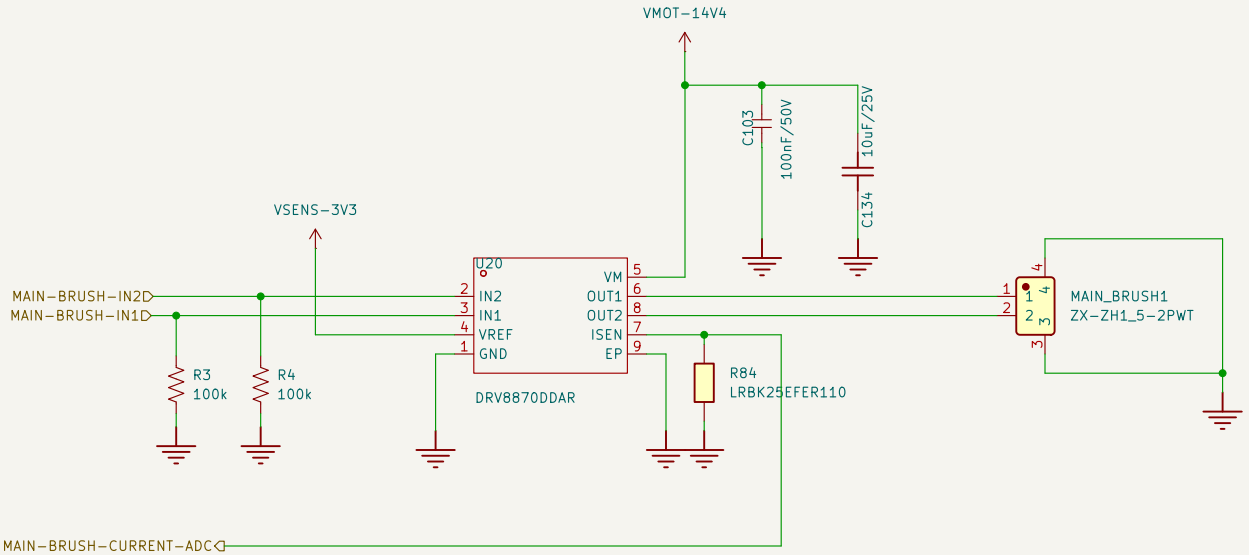
ANTI FALL SENSORs



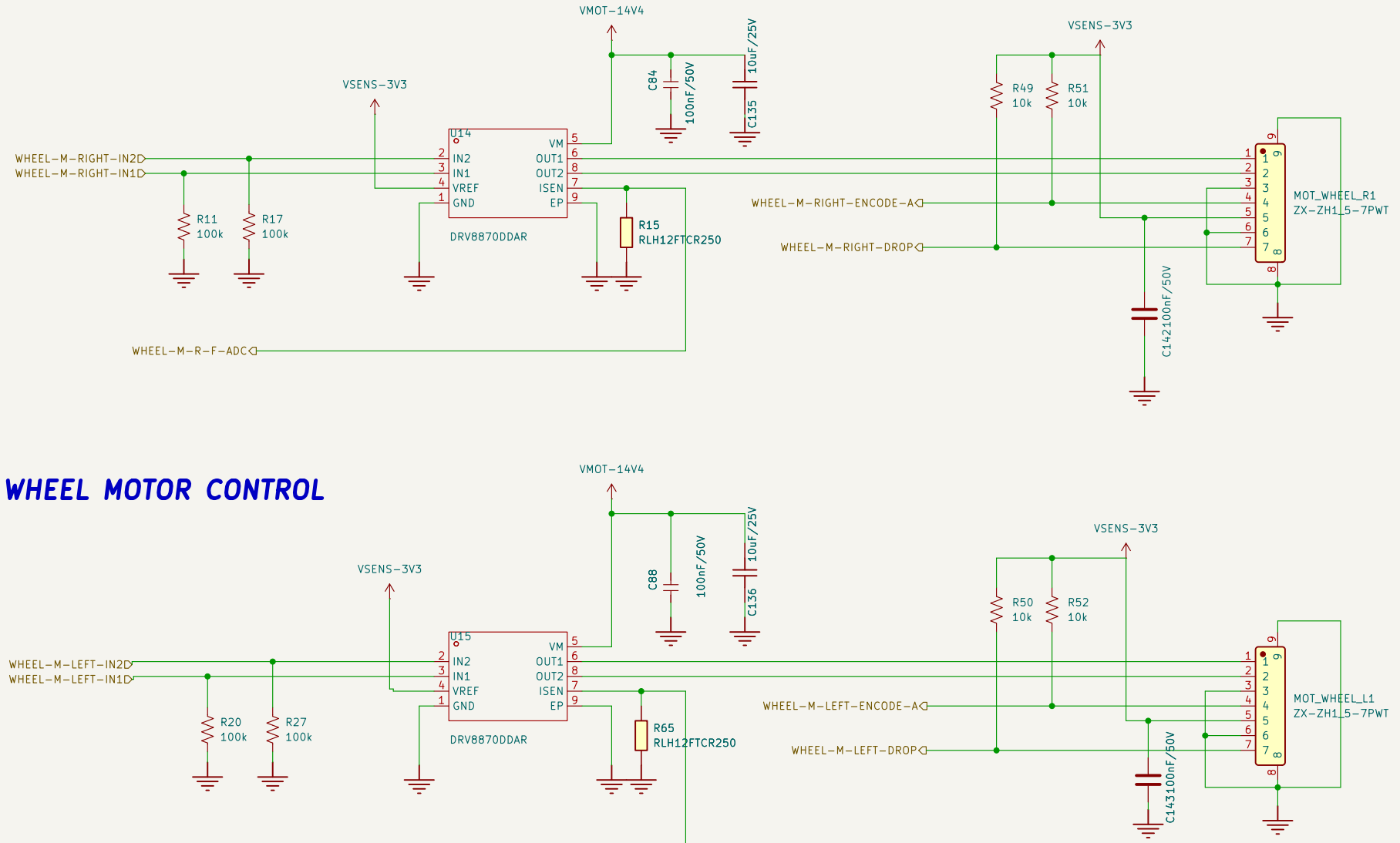
Cliff sensors: 6 boards (kicad/cliff-sensor), SH 4-pin vertical: 1 VSENS-5V-USB (LED, 150R on the board),
2 CLIFF-LED-K (Q35 sinks all 6 LEDs), 3 phototransistor collector (47k pull-up), 4 GND.
Modulate CLIFF_LED_EN and read each channel LED-on minus LED-off to cancel ambient light.
ADC: LF PA3, RF PA1, LR PA2, RR PA0, LM PD9, RM PE8 (LM/RM via global labels).

Bumpers: 2x ITR9606 slotted photo-interrupters on the main board (~11 mm tall, set by the bumper
mechanism). Emitters in series from VSENS-5V-USB, switched by BUMPER_LED_EN (high = on, 100k pull-down).
BUMP-x-Q+ reads low when light passes the slot (R44/R48 47k pull-ups).

Sheet: /CLIFF/	
File: CLIFF.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 19/15



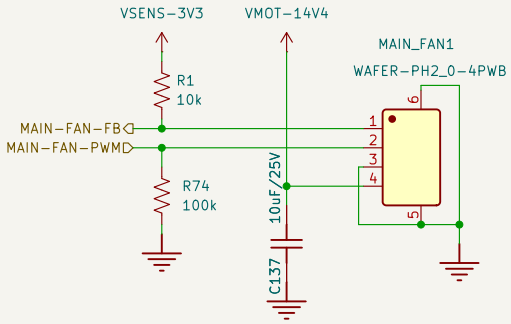
Sheet: /MAIN-BRUSH MOTOR/	
File: MAIN-BRUSH-MOTOR .kicad_sch	
Title:	
Size: USLetter Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 22/15



Sheet: /WHEEL MOTOR/
File: WHEEL-MOTORs .kicad_sch

Title:

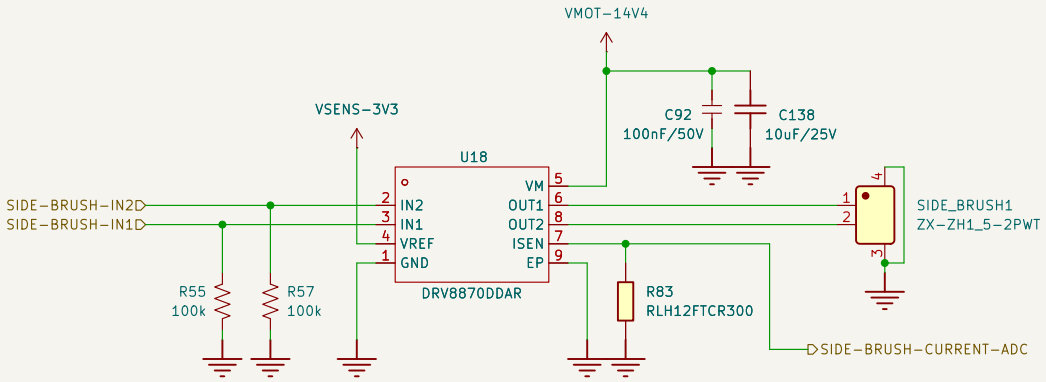
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 23/15



- feedback signal is open collector
- set PWM LOW to turn fan OFF

Sheet: /MAIN-FAN/		
File: MAIN-FAN .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 24/15

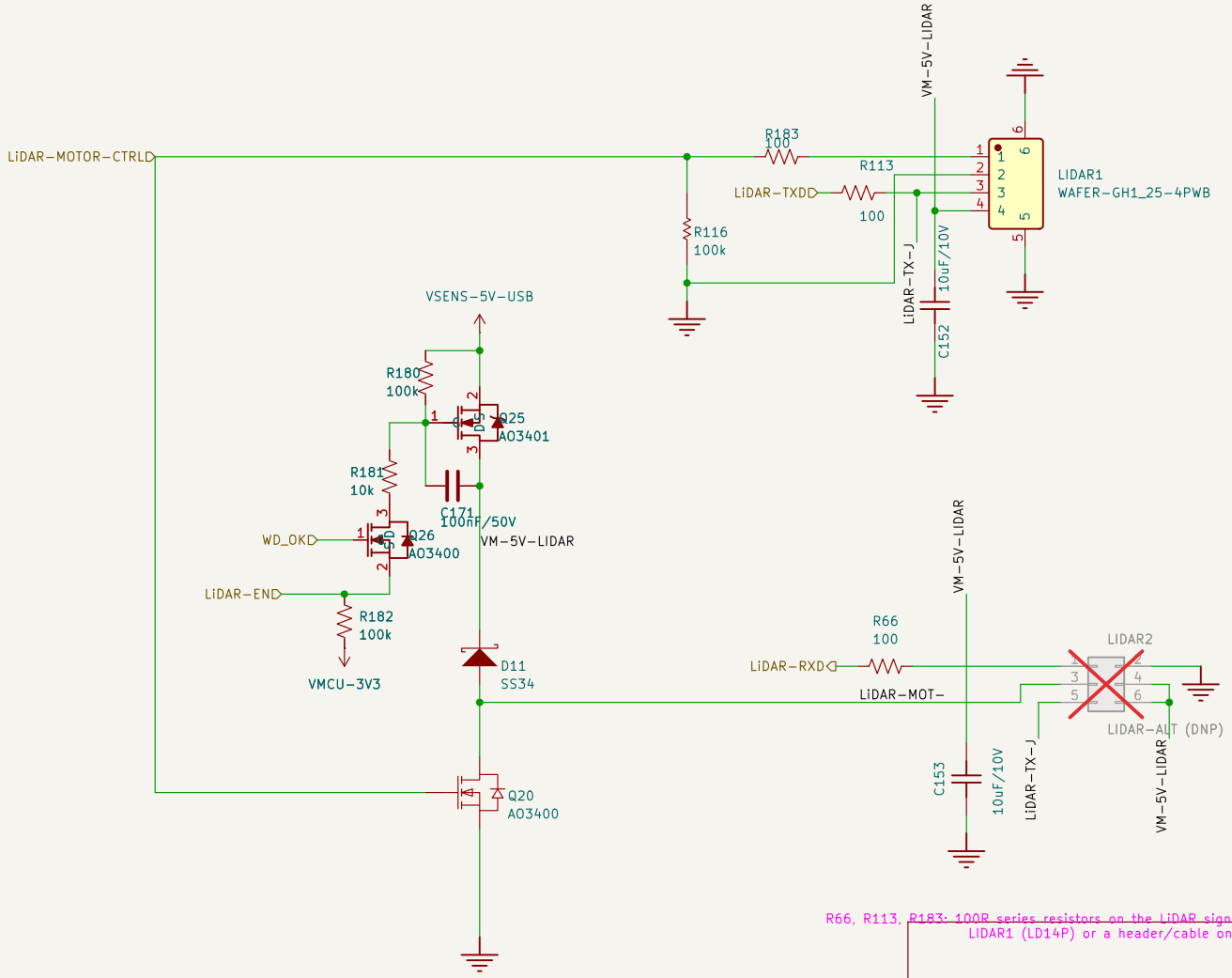
SIDE-BRUSH MOTOR CONTROL



Sheet: /SIDE-BRUSH/	
File: SIDE-BRUSH-MOTORs .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 25/15

VM−5V−LIDAR powers the whole LiDAR (laser included) for any model. On only while WD_OK is high
AND LiDAR−EN is low (Q26 source−switched AND). R181/C171: ~1.5 ms soft start, ~30 ms soft stop.
LIDAR2: 2x3 0.1in through−hole pads (not fitted) for other LiDARs. Pin 6 = VM−5V−LIDAR for LiDARs
that need a separate motor supply (motor between pins 6 and 3). Pin 5 = the LiDAR TX, shared with LIDAR1 pin 3.

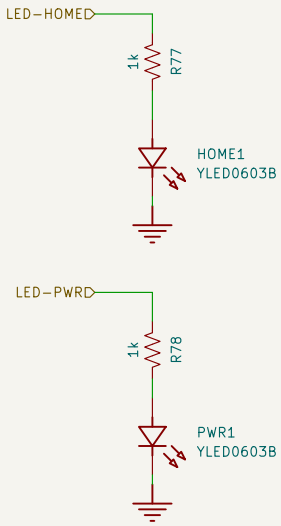
LiDAR

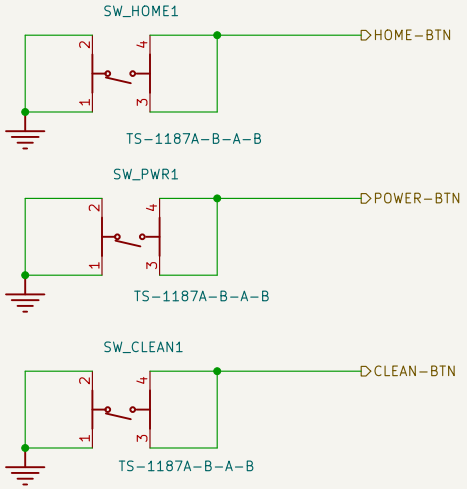


R66, R113, R183: 100R series resistors on the LiDAR signals. Only one LiDAR is fitted:
LIDAR1 (LD14P) or a header/cable on LIDAR2.

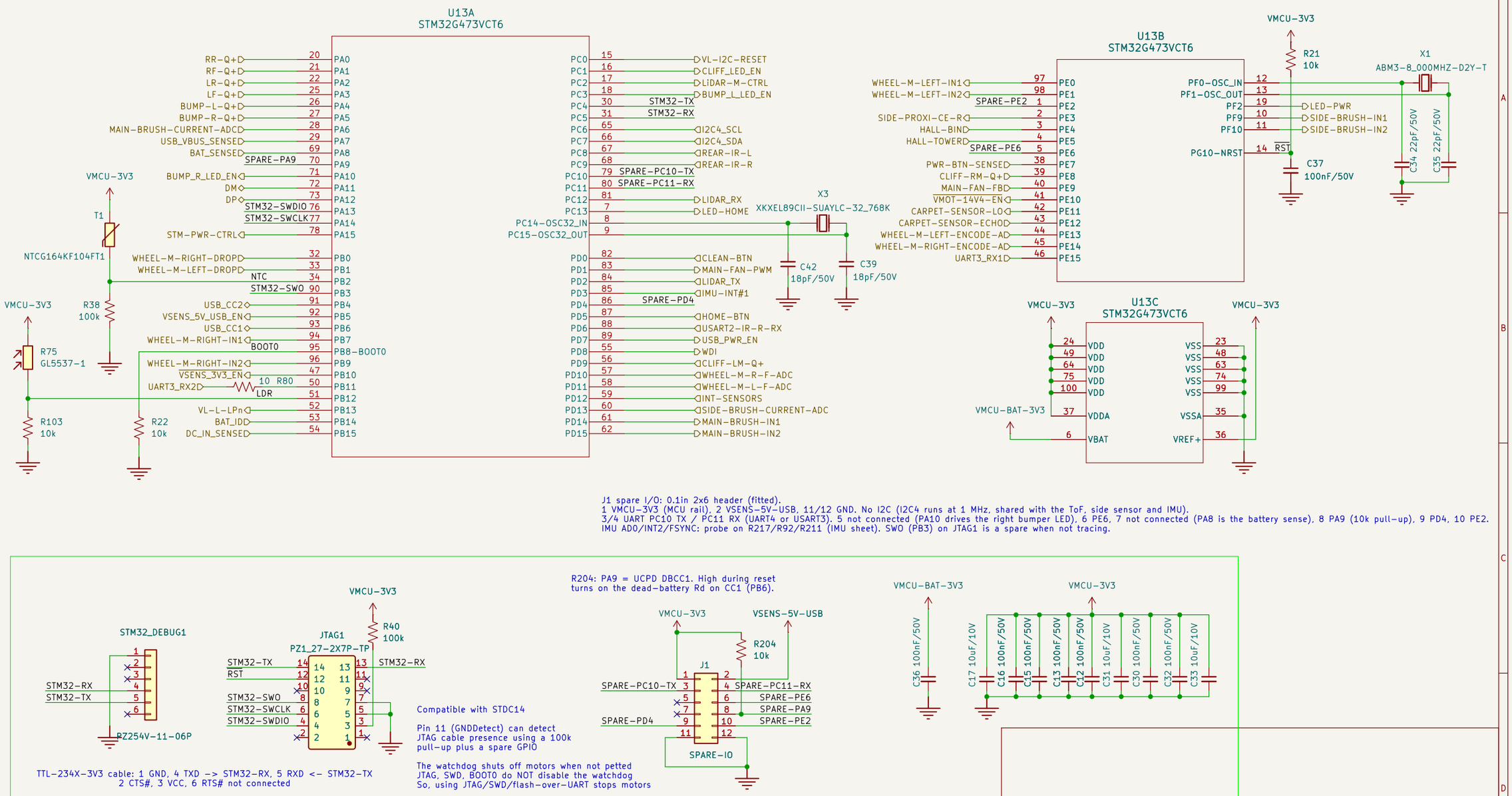
Sheet: /LiDAR/		
File: LiDAR.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 26/15

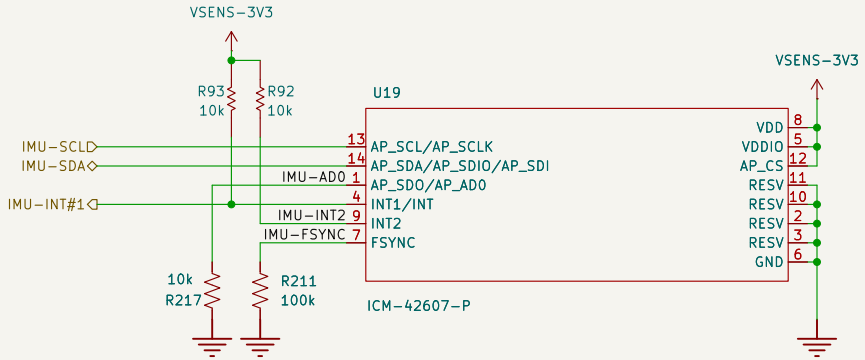
LEDs AND BUTTONs



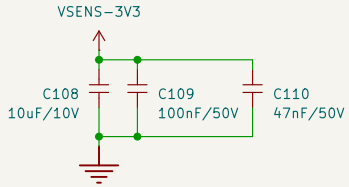


Sheet: /BUTTON-LED/ File: BUTTON-LEDs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 28/15


U13A
STM32G473VCT6
RR-Q+ 20 PA0
RF-Q+ 21 PA1
LR-Q+ 22 PA2
LF-Q+ 25 PA3
BUMP-L-Q+ 26 PA4
BUMP-R-Q+ 27 PA5
MAIN-BRUSH-CURRENT-ADC 28 PA6
USB_VBUS_SENSED 29 PA7
BAT_SENSED 69 PA8
SPARE-PA9 70 PA9
BUMP_R_LED_EN 71 PA10
DM 72 PA11
DP 73 PA12
STM32-SWDIO 76 PA13
STM32-SWCLK 77 PA14
STM-PWR-CTRL 78 PA15
WHEEL-M-RIGHT-DROP 32 PB0
WHEEL-M-LEFT-DROP 33 PB1
NTC 34 PB2
STM32-SWO 90 PB3
USB_CC2 91 PB4
VSENS_5V_USB_EN 92 PB5
USB_CC1 93 PB6
WHEEL-M-RIGHT-IN1 94 PB7
BOOT0 95 PB8-BOOT0
WHEEL-M-RIGHT-IN2 96 PB9
VSENS_3V3_EN 47 PB10
UART3_RX2 10 R80 LDR 50 PB11
51 PB12
VL-L-LPn 52 PB13
BAT_ID 53 PB14
DC_IN_SENSED 54 PB15
PC0 15 VL-I2C-RESET
PC1 16 CLIFF_LED_EN
PC2 17 LIDAR-M-CTRL
PC3 18 BUMP_L_LED_EN
PC4 30 STM32-TX
PC5 31 STM32-RX
PC6 65 I2C4_SCL
PC7 66 I2C4_SDA
PC8 67 REAR-IR-L
PC9 68 REAR-IR-R
PC10 79 SPARE-PC10-TX
PC11 80 SPARE-PC11-RX
PC12 81 LIDAR_RX
PC13 7 LED-HOME
PC14-OSC32_IN 8
PC15-OSC32_OUT 9
PD0 82 CLEAN-BTN
PD1 83 MAIN-FAN-PWM
PD2 84 LIDAR_TX
PD3 85 IMU-INT#1
PD4 86 SPARE-PD4
PD5 87 HOME-BTN
PD6 88 USART2-IR-R-RX
PD7 89 USB_PWR_EN
PD8 55 WDI
PD9 56 CLIFF-LM-Q+
PD10 57 WHEEL-M-R-F-ADC
PD11 58 WHEEL-M-L-F-ADC
PD12 59 INT-SENSORS
PD13 60 SIDE-BRUSH-CURRENT-ADC
PD14 61 MAIN-BRUSH-IN1
PD15 62 MAIN-BRUSH-IN2
VMCU-3V3
T1
NTCG164KF104FT1
R38 100k
R75 GL5537-1
R103 10k
R22 10k
X3
XKXEL89CII-SUAYLC-32_768K
C42 18pF/50V
C39 18pF/50V
U13B
STM32G473VCT6
WHEEL-M-LEFT-IN1 97 PE0
WHEEL-M-LEFT-IN2 98 PE1
SPARE-PE2 1 PE2
SIDE-PROXI-CE-R 2 PE3
HALL-BIND 3 PE4
HALL-TOWERD 4 PE5
SPARE-PE6 5 PE6
PWR-BTN-SENSED 38 PE7
CLIFF-RM-Q+ 39 PE8
MAIN-FAN-FB 40 PE9
VMOT-14V4-EN 41 PE10
CARPET-SENSOR-LO 42 PE11
CARPET-SENSOR-ECHO 43 PE12
WHEEL-M-LEFT-ENCODE-AD 44 PE13
WHEEL-M-RIGHT-ENCODE-AD 45 PE14
UART3_RX1 46 PE15
VMCU-3V3
R21 10k
X1
ABM3-8_000MHZ-D2Y-T
C34 22pF/50V
C35 22pF/50V
PF0-OSC_IN 12
PF1-OSC_OUT 13
PF2 19 LED-PWR
PF9 10 SIDE-BRUSH-IN1
PF10 11 SIDE-BRUSH-IN2
PG10-NRST 14 RST
C37 100nF/50V
U13C
STM32G473VCT6
VMCU-3V3
VMCU-BAT-3V3
24 VDD
49 VDD
64 VDD
75 VDD
100 VDD
37 VDDA
6 VBAT
23 VSS
48 VSS
63 VSS
74 VSS
99 VSS
35 VSSA
36 VREF+
J1 spare I/O: 0.1in 2x6 header (fitted).
1 VMCU-3V3 (MCU rail), 2 VSENS-5V-USB, 11/12 GND. No I2C (I2C4 runs at 1 MHz, shared with the ToF, side sensor and IMU).
3/4 UART PC10 TX / PC11 RX (UART4 or USART3). 5 not connected (PA10 drives the right bumper LED), 6 PE6, 7 not connected (PA8 is the battery sense), 8 PA9 (10k pull-up), 9 PD4, 10 PE2.
IMU AD0/INT2/FSYNC: probe on R217/R92/R211 (IMU sheet). SWO (PB3) on JTAG1 is a spare when not tracing.
STM32_DEBUG1
STM32-RX
STM32-TX
PZ254V-11-06P
VMCU-3V3
R40 100k
JTAG1
PZ1_27-2X7P-TP
STM32-TX
RST
STM32-SWO
STM32-SWCLK
STM32-SWDIO
STM32-RX
Compatible with STDC14
Pin 11 (GNDDetect) can detect JTAG cable presence using a 100k pull-up plus a spare GPIO
The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors
TTL-234X-3V3 cable: 1 GND, 4 TXD -> STM32-RX, 5 RXD <- STM32-TX
2 CTS#, 3 VCC, 6 RTS# not connected
R204: PA9 = UCPD DBCC1. High during reset turns on the dead-battery Rd on CC1 (PB6).
VMCU-3V3
VSENS-5V-USB
R204 10k
J1
SPARE-PC10-TX
SPARE-PD4
SPARE-PC11-RX
SPARE-PE6
SPARE-PA9
SPARE-PE2
SPARE-IO
VMCU-BAT-3V3
VMCU-3V3
C36 100nF/50V
C17 10uF/10V
C16 100nF/50V
C15 100nF/50V
C13 100nF/50V
C12 100nF/50V
C31 10uF/10V
C30 100nF/50V
C32 100nF/50V
C33 10uF/10V
Sheet: /MCU-STM32/
File: STM32G473.kicad_sch
Title:
Size: USLegal
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 30/15

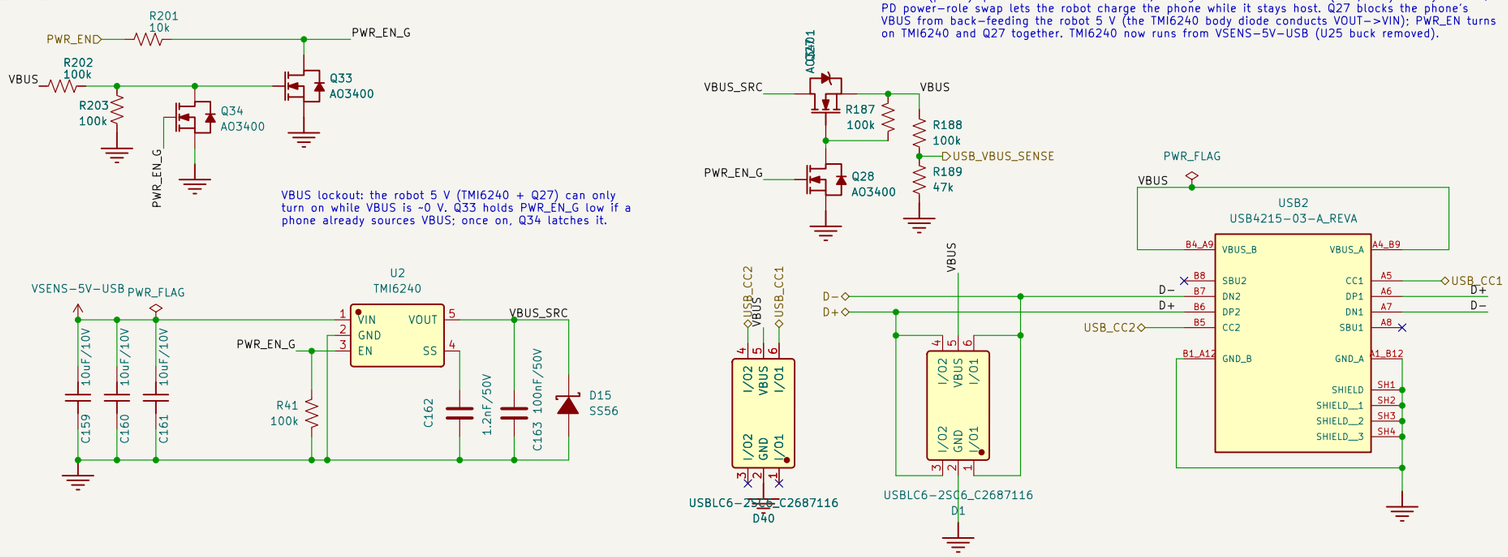


IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

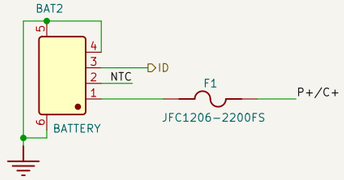


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Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 32/15

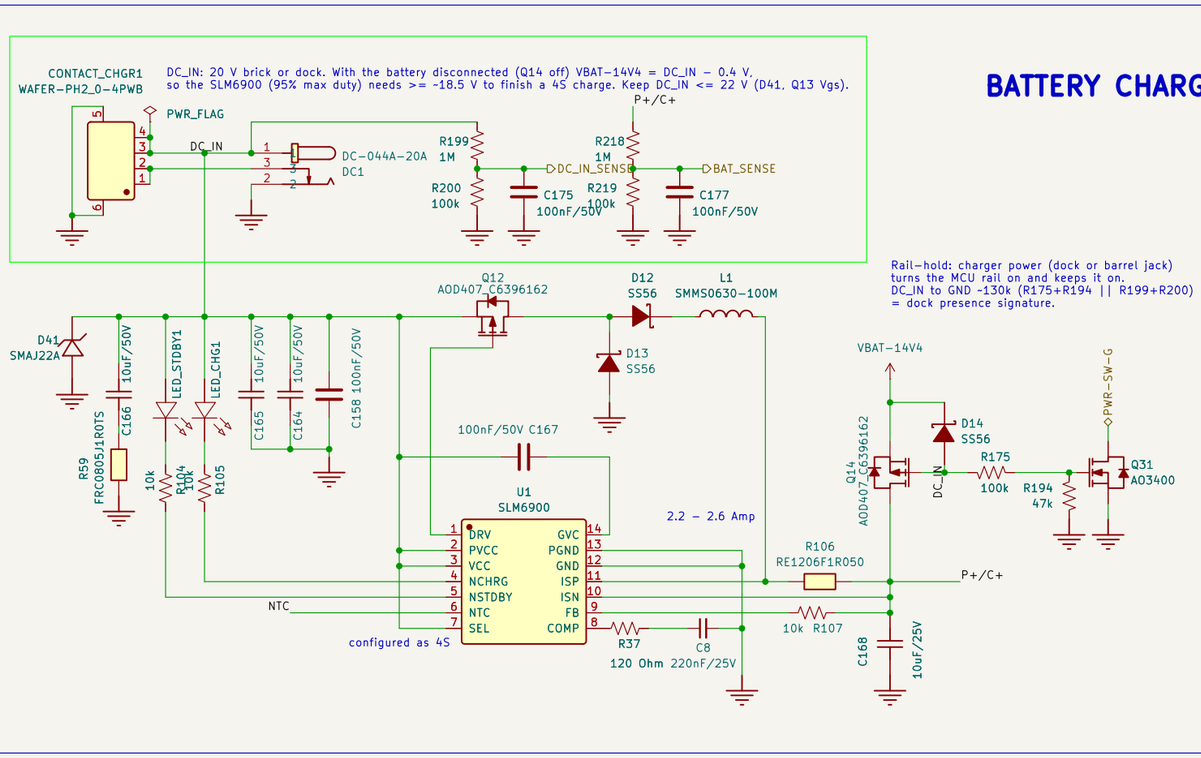
USB-C



BATTERY CONNECTOR



BATTERY CHARGER



Sheet: /BATTERY-CHARGER/		
File: Battery-Charger.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 33/15